Lyntra Data

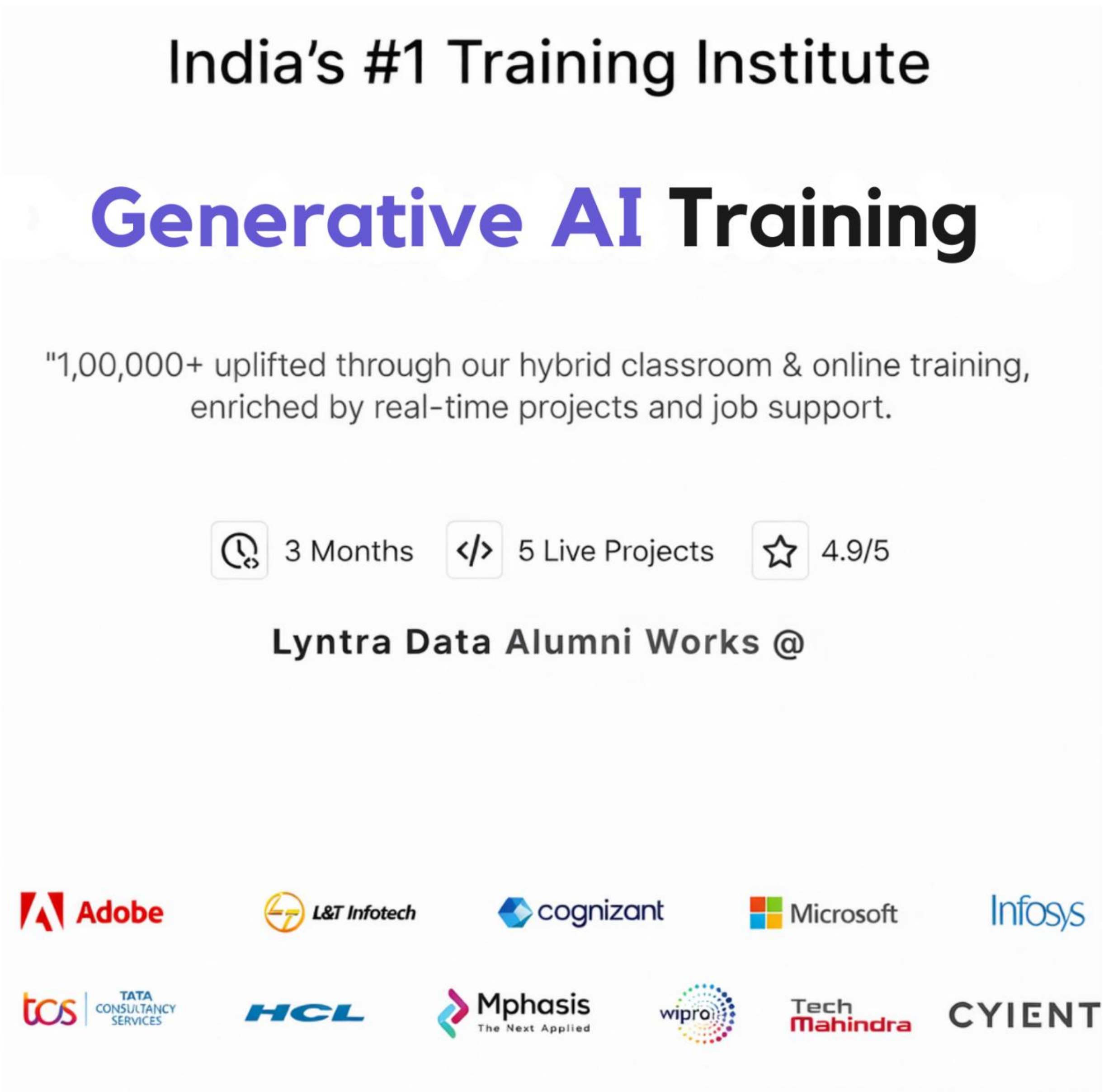
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TABLE OF CONTENTS

Fundamentals of IT & AI

Python for AI

Foundations of AI

Transformers

Module 4: Transformer Variants and Implementation

Module 5: Applications, LLMs & Ethics

Gen AI

Module 1: Foundations of Generative AI

Module 2: Prompt Engineering & Interaction Design

Module 3: Transformers and Large Language Models

Module 4: Tools, Frameworks & Automation

Module 5: Applications and Deployment

AI Agents

Module 1: Fundamentals of AI Agents

Module 2: Reasoning, Decision-Making & Learning

Module 3: Multi-Agent Systems

Module 4: Frameworks & Tools for Agent Development

Module 5: Building & Deploying Scalable Agents

RAG & Vector Databases

Module 1: Introduction to RAG and Vector Databases

Module 2: Embeddings and Semantic Search

Module 3: Vector Database Platforms

Module 4: Integrating RAG with LLMs

Module 5: Advanced RAG Applications & Deployment

AI Agent Frameworks

Module 1: LangChain and Context Management

Module 2: LangGraph and Stateful Agents

Module 3: CrewAI Framework

Module 4: AutoGen and Emerging Frameworks

Module 5: Model Context Protocol (MCP) and Interoperability

AI Agents in Production

Module 1: Development Tools & Testing

Module 2: Security, Safety & Compliance

Module 3: Deployment Architecture

Module 4: Scalability & Performance

Module 5: Monitoring, Maintenance & Governance

Fundamentals of IT & AI

Module 1: The Software Application Life Cycle

- What is an Application?
- Types of Applications
- Web Application Fundamentals
- Web Technologies:
  - **Frontend:** HTML, CSS, JavaScript, React
  - **Backend:** Python, Java, Node.js
  - **Databases:** SQL (MySQL, PostgreSQL), NoSQL (MongoDB)
- Software Development Life Cycle (SDLC)
  - **Phases:** Planning, Analysis, Design, Implementation (Coding), Testing, Deployment, Maintenance.
- Application Development Methodologies
  - **Agile:** Core principles, Scrum, Kanban
  - **Waterfall**

Module 2: Data Fundamentals

- What is Data?
- Types of Data
- Data Storage
- Data Analysis
- Data Engineering
- Data Science

Module 3: Computing and the Cloud

- The Importance of Computing Power
- Key Computing Technologies:
  - CPU (Central Processing Unit)
  - GPU (Graphics Processing Unit)
- Cloud Computing:
  - What is the Cloud?
  - Cloud Service Models:
    - IaaS (Infrastructure as a Service)
    - PaaS (Platform as a Service)
    - SaaS (Software as a Service)

Module 4: Introduction to AI and Generative AI

- What is Artificial Intelligence (AI)?
- How AI Works?
- Key Concepts:
  - Machine Learning (ML)
  - Deep Learning (DL)
- Generative AI:

- What is Generative AI?
- Examples: Large Language Models (LLMs), image generation models.

- AI in Everyday Learning

Module 5: Real-World Applications of Technology

- Customer Relationship Management (CRM)
- Human Resource Management Systems (HRMS)
- Retail & E-Commerce
- Healthcare

Python for AI

Module 1: Python Foundations

- Introduction to Python and its real-world applications
- Installation, setup, and environment configuration
- Writing and executing Python scripts
- Syntax, keywords, and identifiers
- Variables, data types, and operators
- Input/output and basic program structure

Module 2: Control Flow and Functions

- Conditional statements: if, elif, else
- Loops: for, while, and control statements (break, continue, pass)
- Functions — definition, arguments, and return values
- Types of arguments: positional, keyword, default, variable-length
- Lambda functions, Map, Filter, and Reduce
- Modular programming and code reusability

Module 3: Working with Data Structures

- Strings — slicing, formatting, and common methods
- Lists, Tuples, Sets, and Dictionaries
- Mutability and immutability concepts
- Nested data structures and comprehension techniques
- Advanced collection methods and iterators
- Frozen sets and when to use them

Module 4: File Handling, Modules, and Exception Management

- File operations: read, write, append, context managers (`with` statement)
- Handling CSV and Excel files
- Creating and importing custom modules and packages
- Exception handling: try, except, finally, else, raise
- Creating and using custom exceptions
- Regular expressions for pattern matching and text validation

Module 5: Object-Oriented and Advanced Python Concepts

- OOP fundamentals: classes, objects, attributes, and methods
- Constructors, destructors, and method types (instance/class/static)
- Inheritance, polymorphism, encapsulation, and abstraction
- Decorators and property methods
- Overview of functional and modular design
- Capstone mini-project combining OOP, file handling, and exception management

Foundations of AI

Module 1: Introduction to Artificial Intelligence

- What is AI? Evolution and core objectives
- Types of AI: Narrow, General, and Super AI
- Key subfields: ML, DL, NLP, Robotics, and Expert Systems
- Real-world applications of AI across industries
- Ethics, bias, and responsible AI foundations

Module 2: Machine Learning Fundamentals

- What is Machine Learning and how it differs from traditional programming
- Categories of ML: Supervised, Unsupervised, Reinforcement Learning
- Core concepts: Features, Labels, Training, Testing, Overfitting, Underfitting
- Common algorithms: Linear/Logistic Regression, K-Means, Decision Trees
- Evaluation metrics: Accuracy, Precision, Recall, F1-Score

Module 3: Deep Learning and Neural Networks

- What is Deep Learning and how it extends ML
- Anatomy of a Neural Network — neurons, layers, activation functions
- Forward and backward propagation (high-level overview)
- Key architectures: CNNs (vision), RNNs (sequence), Transformers (foundation for GenAI)
- Frameworks overview: TensorFlow, PyTorch, Keras

Module 4: Natural Language Processing (NLP) Essentials

- Understanding language data — text preprocessing and tokenization
- Word embeddings: Word2Vec, GloVe, and contextual embeddings
- Core NLP tasks: Text classification, Named Entity Recognition, Sentiment Analysis

- Sequence models: RNN, LSTM, GRU, Transformer introduction
- NLP applications in chatbots, summarization, and Q&A systems

Module 5: AI Integration & Readiness for GenAI

- From traditional AI to Generative AI — key differences
- How data, models, and compute enable GenAI
- Role of large language models (LLMs) and foundational models
- Responsible AI and data governance principles
- Preparing development environment and mindset for GenAI & AI agents

Transformers

Module 1: Neural Networks Refresher & Sequence Modeling

- Overview of perceptrons, feedforward, and deep neural networks
- Limitations of traditional sequence models (RNNs, LSTMs, GRUs)
- Understanding sequential data and context representation
- Vanishing gradient problem and need for attention mechanisms
- Transition from recurrent to attention-based architectures

Module 2: Attention Mechanism

- Concept and intuition behind attention
- Queries, Keys, and Values — how attention focuses on relevant context
- Scaled dot-product attention — computation and visualization
- Self-attention vs. cross-attention
- Solving long-term dependencies using attention

Module 3: Transformer Core Architecture

- Encoder–decoder architecture overview
- Multi-head attention and parallel information processing
- Feed-forward layers, residual connections, and layer normalization
- Positional encoding to retain sequence order
- Comparison between RNNs and Transformers

Module 4: Transformer Variants and Implementation

- Encoder-only, decoder-only, and encoder–decoder architectures
- Overview of BERT, GPT, and T5 models
- Tokenization and embedding strategies
- Masking techniques (padding, look-ahead)
- Implementing a transformer model using frameworks (PyTorch/Hugging Face)

Module 5: Applications, LLMs & Ethics

- From Transformers to Large Language Models (LLMs)
- Fine-tuning and transfer learning for NLP tasks
- Common applications: translation, summarization, text generation
- Limitations, biases, and ethical considerations in LLMs
- Future trends and research directions in transformer-based AI

Gen AI

Module 1: Foundations of Generative AI

- What is Generative AI and how it differs from traditional AI
- Evolution: Rule-Based * Deep Learning * Transformers * LLMs
- Core architectures: GANs, VAEs, Diffusion Models, LLMs

- Popular tools: ChatGPT, Claude, Midjourney, DALL·E, Copilot
- Ethical and responsible AI principles

Module 2: Prompt Engineering & Interaction Design

- Prompt design principles: Zero-shot, Few-shot, Instruction-based
- Advanced prompting: Chain-of-Thought, self-consistency, iterative refinement
- Context management and multi-turn prompting
- Evaluating and debugging model outputs
- Hands-on: optimizing prompts for text, image, and code generation

Module 3: Transformers and Large Language Models

- Architecture deep dive: Self-Attention, Multi-Head Attention
- Encoder vs Decoder vs Encoder-Decoder models
- Popular families: GPT, T5, LLaMA, Gemini
- Pre-training, fine-tuning, embeddings, and tokenization
- Model limitations and evaluation (bias, hallucination, cost)

Module 4: Tools, Frameworks & Automation

- Libraries: Hugging Face Transformers, OpenAI API, Cohere
- Workflow automation using n8n or Zapier
- Model experimentation in Colab, Jupyter, or local setup
- Quality metrics and output monitoring
- Responsible content generation and compliance

Module 5: Applications and Deployment

- AI content generation: text, image, code, and multimedia
- Cloud vs on-premise deployment of GenAI models
- Integrating APIs into products and web apps

- Case studies: Copilot, customer support bots, creative AI
- Capstone: build and deploy a mini GenAI-based application

AI Agents

Module 1: Fundamentals of AI Agents

- Definition, characteristics, and life cycle of AI agents
- Autonomous vs semi-autonomous agents
- Reactive, goal-based, deliberative, and hybrid agents
- Applications: automation, robotics, digital assistants

Module 2: Reasoning, Decision-Making & Learning

- Knowledge representation: rules, ontologies, graphs
- Decision-making: utility-based and goal-oriented planning
- Learning paradigms: supervised, unsupervised, reinforcement
- Feedback loops and environment sensing
- Mini project: build a simple task-based reasoning agent

Module 3: Multi-Agent Systems

- Coordination, communication, and negotiation strategies
- Collaboration and distributed intelligence
- Conflict resolution and resource allocation
- Simulation environments for agent teamwork
- Case study: multi-agent workflow automation

Module 4: Frameworks & Tools for Agent Development

- Overview: CrewAI, LangChain, LangGraph, AutoGen
- Workflow orchestration tools (n8n, Zapier)
- Connecting agents to APIs, data sources, and external services

- Monitoring, observability, and agent lifecycle management

Module 5: Building & Deploying Scalable Agents

- Designing goal-oriented and collaborative agent systems
- Cloud deployment & scaling multi-agent systems
- Real-world use cases: customer support, business automation
- Ethics, safety, and reliability of autonomous agents
- Capstone: Deploy an AI agent workflow using CrewAI or LangChain

RAG & Vector Databases

Module 1: Introduction to RAG and Vector Databases

- What is RAG (Retrieval-Augmented Generation)?
- How RAG differs from standard LLM use
- Overview of vector databases and embeddings
- Applications: QA systems, document search, knowledge-grounded chatbots

Module 2: Embeddings and Semantic Search

- Understanding embeddings: word, sentence, and document vectors
- Generating embeddings with OpenAI, Hugging Face, and Cohere
- Similarity measures: cosine similarity, Euclidean distance
- Semantic search vs keyword search

Module 3: Vector Database Platforms

- Popular vector DBs: Pinecone, Weaviate, Milvus, FAISS
- Indexing and retrieval methods
- CRUD operations on vector data
- Scaling vector DBs for production

Module 4: Integrating RAG with LLMs

- Connecting LLMs to vector DBs for retrieval-augmented responses
- Prompting strategies for RAG
- Handling long documents, multi-document retrieval, and context windows
- Example pipelines: LangChain + Vector DB + LLM

Module 5: Advanced RAG Applications & Deployment

- Hybrid search: combining embeddings with traditional search
- RAG in multi-agent systems
- Deployment considerations: latency, caching, and real-time updates
- Monitoring and evaluating RAG performance

AI Agent Frameworks

Module 1: LangChain and Context Management

- LangChain fundamentals: chains, memory, tools, agents
- Prompt templates and conversation flow
- Integrating external APIs and document loaders
- LangSmith and LangServe for monitoring

Module 2: LangGraph and Stateful Agents

- Understanding LangGraph architecture
- Building stateful, event-driven agent systems
- Handling context, memory, and transitions
- Debugging and visualization of agent graphs

Module 3: CrewAI Framework

- Multi-agent collaboration with CrewAI
- Role assignment, task orchestration, and hierarchy management

- Integrating CrewAI with LangChain and APIs
- Hands-on: building a multi-role task automation agent

Module 4: AutoGen and Emerging Frameworks

- Introduction to AutoGen and other rising frameworks
- LLM-to-LLM communication and collaboration
- Building self-improving agent loops
- Model Context Protocol (MCP) introduction

Module 5: Model Context Protocol (MCP) and Interoperability

- What is MCP and why it matters
- Standardizing communication between AI systems
- Security, governance, and interoperability considerations
- Case study: integrating multiple agent frameworks via MCP

AI Agents in Production

Module 1: Development Tools & Testing

- Best practices for agent code structure and versioning
- Testing strategies: simulation vs live environment
- Debugging agent behaviors and response tracing
- Using playgrounds and sandboxes for safe experimentation

Module 2: Security, Safety & Compliance

- Threat modeling and adversarial risks
- Data privacy and secure API access
- Responsible agent policies and guardrails
- Regulatory frameworks (GDPR, SOC2, ISO27001)

Module 3: Deployment Architecture

- Cloud-native vs on-premise agent deployment
- Microservices and containerization (Docker, Kubernetes)
- CI/CD pipelines for AI workflows
- Integrating agents into enterprise applications

Module 4: Scalability & Performance

- Load balancing and distributed execution
- Monitoring agent throughput and latency
- Model caching and performance optimization
- Cost efficiency and resource allocation

Module 5: Monitoring, Maintenance & Governance

- Continuous improvement loops
- Observability dashboards (Prometheus, Grafana)
- Retraining and upgrading deployed agents
- Ethical operations and transparency reports
- Capstone: deploy and monitor a production-grade AI agent system

Tools

AI-powered coding assistant for efficiency and automation.

Git is a distributed version control system for tracking code changes.

Jenkins is an open-source tool for automating CI/CD pipelines.

Kubernetes automates deployment, scaling, and management of containers.

AI model for generating and assisting with code and text.

Google's AI model for assisting in code generation, debugging, and research.

Power BI is a business analytics tool for visualizing data insights.

Version control tools for managing and collaborating on test automation scripts.

Python powers everything from web apps to AI.

Core framework for building autonomous AI agents and workflows.

Tools

Linux is an open-source operating system for servers, desktops, and devices.

Nexus is a repository manager for managing build artifacts and dependencies.

Slack is a messaging platform for teams and workplace collaboration.

Trello is a visual tool for organizing tasks and projects.

SQL is a language for managing and querying relational databases.

Lightweight and customizable code editor with Python support.

Interactive coding environment for testing DSA concepts.

One of the best platforms for solving DSA problems.

Step-by-step visualization of code execution.